PROJECT SYNOPSIS

CryptoGuard

A Measured Password and File-Security Workbench

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Course	Design Project – I · Semester III · 2024–25
Domain	Applied cryptography · password security
Repository	github.com/harshacreate/CryptoGuard

Introduction

Password-strength meters assume an attacker speed that in reality depends on how the server stores the password. CryptoGuard is a local web workbench for password analysis, secure generation, authenticated file encryption, integrity checking and a bounded cracking demonstration, extended with a benchmark that measures storage-scheme cost on the host.

Problem Statement

Users and students cannot see how much the storage scheme (MD5 vs PBKDF2 vs scrypt) changes an attacker's guess rate, and many online checkers require sending secrets to third parties.

Objectives

- Explainable strength analysis with stated assumptions
- CSPRNG password and passphrase generation
- AES-256-GCM file encryption with PBKDF2 key derivation
- Bounded John the Ripper demonstration
- Measured KDF benchmark expressed in equivalent password characters
- Hardened, tested, containerised delivery

Proposed Methodology

Layered Flask application with pure-Python domain modules, safe subprocess adapters for John the Ripper, hashid and OpenSSL, and a single-page UI. The benchmark times each scheme for a fixed budget and converts the slowdown k into log(k)/log(c) characters for alphabet size c. Security controls: CSP and OWASP headers, input bounds, safe logging, rate limits, non-root container.

Tools and Technologies

Python 3.12, Flask, cryptography (OpenSSL), John the Ripper, hashid, Docker, pytest, GitHub Actions.

Expected Outcomes

- Working local application with six tools
- Reproducible measurement of KDF cost on the host
- Automated unit, integration and security tests
- Report, presentation and demonstration

References

- NIST SP 800-63B; NIST SP 800-132; NIST SP 800-38D
- RFC 8018 (PBKDF2); RFC 7914 (scrypt)
- Bonneau, IEEE S&P 2012; Wheeler, USENIX Security 2016
- OWASP Password Storage Cheat Sheet